

Rene Gueye

Data Engineer

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Summary

Data Engineer with 4+ years of experience designing and operating large-scale ETL/ELT pipelines, cloud-native architectures, and ML-enablement platforms on Microsoft Azure and Databricks. Skilled in Python, PySpark, SQL, Delta Lake, Unity Catalog, and MLOps workflows (MLflow), with a track record of improving forecast model performance by 30%, increasing the forecasting perimeter, and maintaining 90%+ pipeline success rates across 20+ production pipelines processing from 500GB to multiple terabytes of data weekly. Strong background in Power BI development and training, data quality automation (Great Expectations), agile delivery, and cross-functional collaboration with data science teams and stakeholders.

Skills

Cloud & Platforms: Azure Databricks, Azure Data Factory, Azure Storage Gen2, Azure Architecture & Governance (Subscriptions, Resource Group, Azure Key Vault), Azure DevOps

Languages: Python, SQL, PySpark, Scala, C, DAX, M

Data Engineering: Apache Spark, Delta Lake, ETL/ELT pipeline design, Great Expectations (data quality)

BI & Automation: Power BI, Power Apps, Power Automate, Logic App, Zoho Analytics

ML / Data Science: MLflow, XGBoost, decision trees, model monitoring & deployment

Databases: PostgreSQL, MySQL, Oracle

Methods & Tools: Agile/Scrum, Git, Jira, Miro, Docker

Experience

Data Engineer

2021 – 2025

Michelin Tires (via CGI)

Clermont-Ferrand, France

Supply Chain context: Built the end-to-end data & ML platform powering tire production forecasting — predicting optimal production volumes to prevent warehouse stockouts and avoid overproduction across manufacturing plants.

- Designed, built, and maintained 20+ production ETL/ELT pipelines in Azure Data Factory, integrating heterogeneous sources (Oracle, Azure Storage Gen2, ERP systems) and processing data volumes ranging from 500GB to multiple terabytes on a daily/weekly basis to feed supply chain forecasting.
- Engineered ML features for demand forecasting models predicting ideal tire production volumes per warehouse, enabling the supply chain team to balance stock levels and avoid both shortages and excess inventory.
- Partnered with the data science team to engineer predictive-model features and built a Databricks/MLflow comparison dashboard to select and promote the best-performing models to production, contributing to a 30% improvement in model accuracy.
- Owned the production release cycle for models and features (new release every 2 months), operating and monitoring 6+ models running in production weekly; doubled the forecasting scope by consistently delivering validated models on schedule.
- Tracked and monitored model performance over time, ensuring forecast drift was detected early and models were retrained before impacting supply chain decisions.
- Orchestrated end-to-end pipeline execution in Azure Data Factory, coordinating Databricks notebook runs and automated data-quality tests (Great Expectations), sustaining a 90% pipeline success rate with rapid failure detection.
- Built a Power BI monitoring tool for real-time pipeline tracking, enabling the team to identify and respond to failures immediately.
- Designed and managed multiple Delta Lake tables governed through Unity Catalog.
- Trained 100+ colleagues on Power BI, data modeling best practices, and effective data storytelling.
- Developed internal tooling to track data quality (Great Expectations) and streamline system migrations.
- Worked in Agile Scrum using Azure DevOps, Git, and Jira — delivering production-ready features in 3-week sprints within a cross-functional data & supply chain team.

Business Intelligence Intern

Cells & Co BioServices

2021 (5 months)

Clermont-Ferrand, France

- Led the migration of BI reporting from Power BI to Zoho Analytics, re-architecting 10+ reports and 15+ tables with complex underlying queries.
- Built new cross-table analytical reports in Zoho Analytics by combining multiple data sources.
- Developed a bonus alerting tool to flag unused BI tools and tables, improving governance and resource efficiency.
- Delivered all migration objectives within the internship timeline.

Education

Master's Degree in Computer Science (Data Engineering)

INP ISIMA

2022

Clermont-Ferrand, France

Bachelor's Degree in Computer Science

University of Clermont Auvergne (UCA)

2020

Clermont-Ferrand, France

Certifications

Databricks Certified Data Engineer Professional — *In progress*

Databricks Certified Data Engineer Associate — *July 2025*